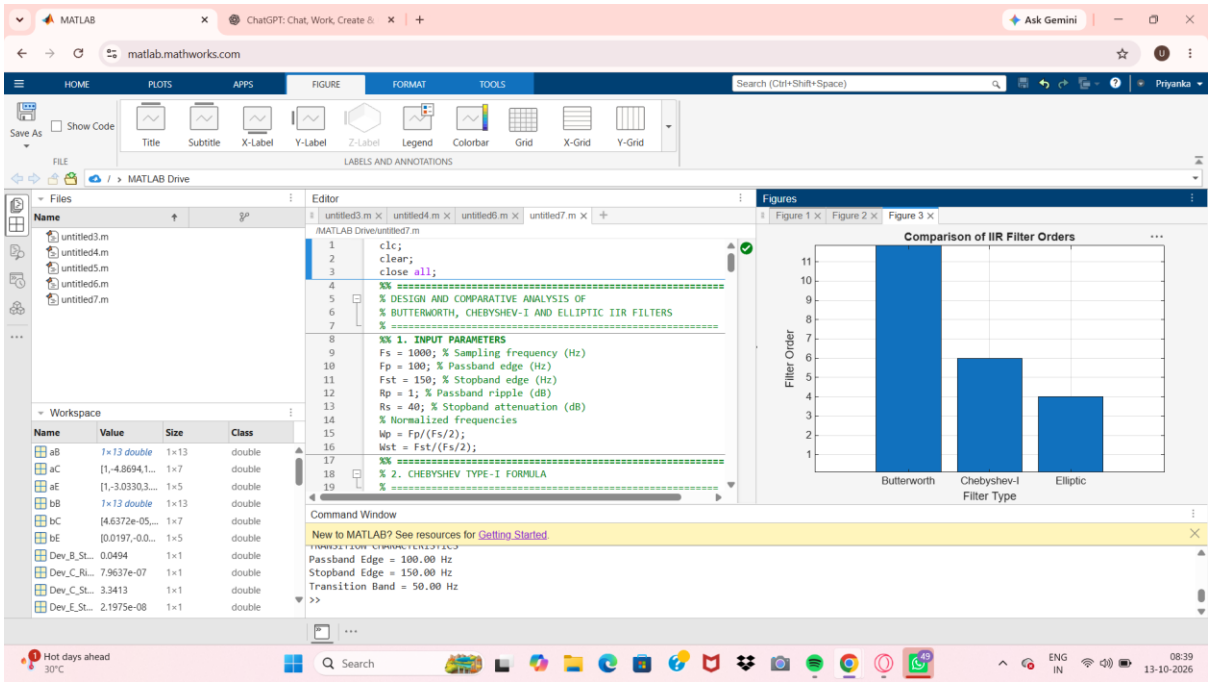
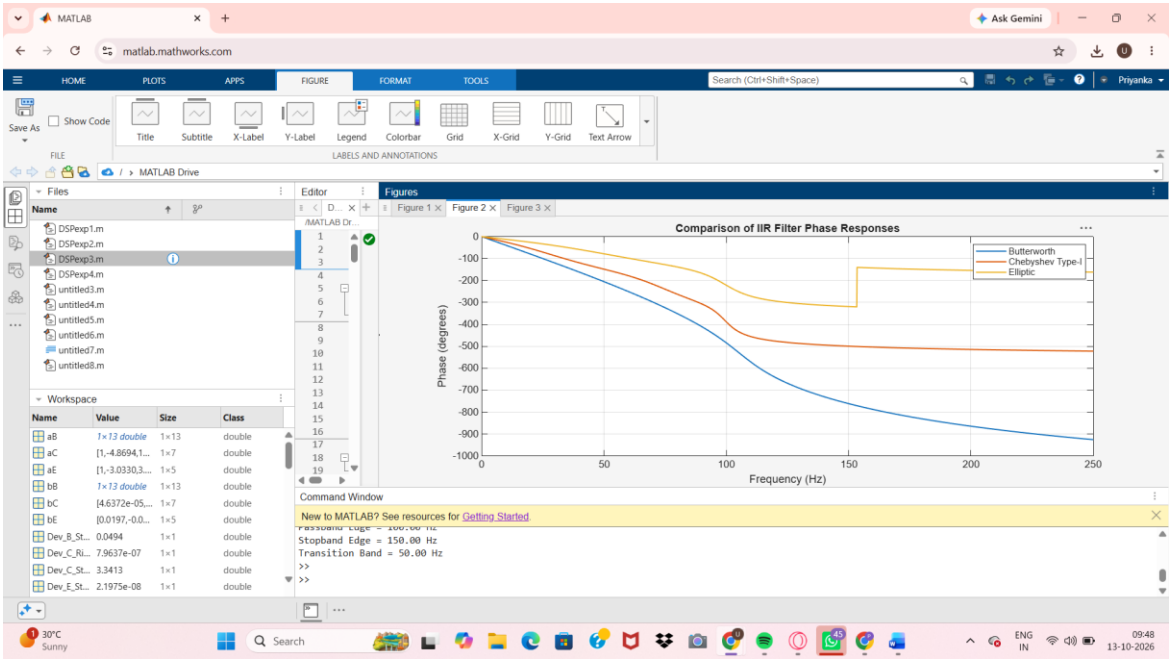
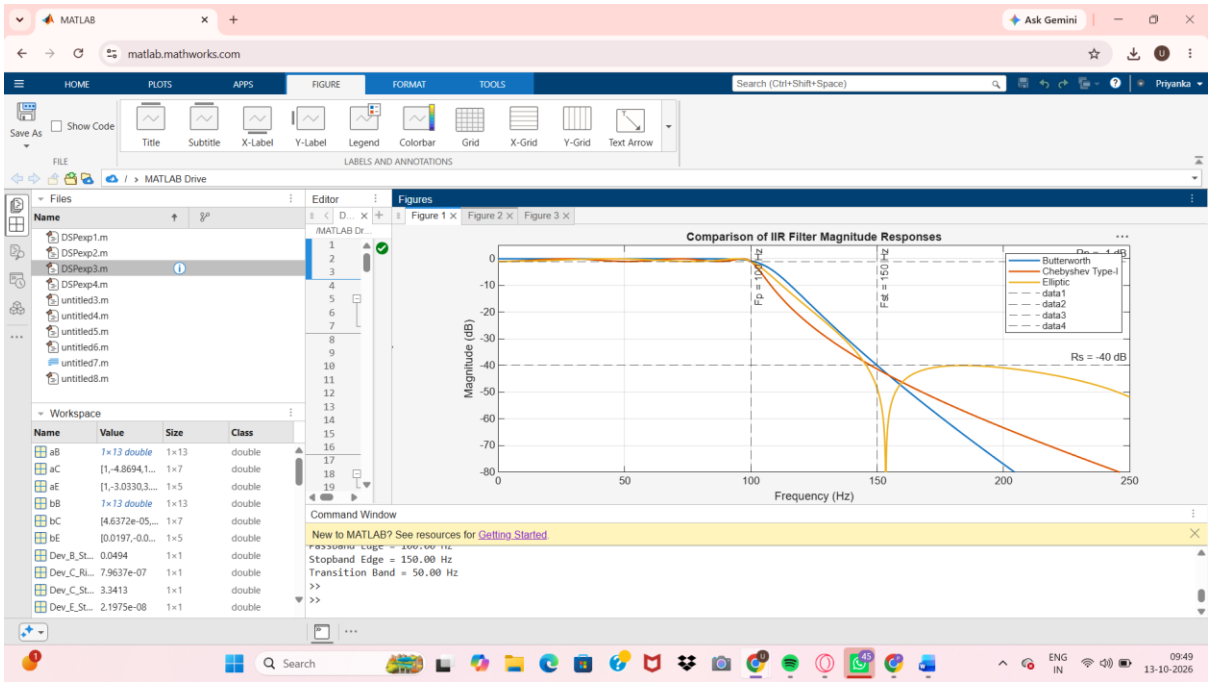

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Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
FILE
LABELS AND ANNOTATIONS
MATLAB Drive
Files
Name
untitled3.m
untitled4.m
untitled5.m
untitled6.m
untitled7.m
Workspace
Name
Value
Size
Class
Editor
untitled3.m
untitled4.m
untitled6.m
untitled7.m
/MATLAB Drive/untitled7.m
clc;
clear;
close all;
%% ==========
% DESIGN AND COMPARATIVE ANALYSIS OF
% BUTTERWORTH, CHEBYSHEV-I AND ELLIPTIC IIR FILTERS
% ==========
%% 1. INPUT PARAMETERS
Fs = 1000; % Sampling frequency (Hz)
Fp = 100; % Passband edge (Hz)
Fst = 150; % Stopband edge (Hz)
Rp = 1; % Passband ripple (dB)
Rs = 40; % Stopband attenuation (dB)
% Normalized frequencies
Wp = Fp/(Fs/2);
Wst = Fst/(Fs/2);
%% ==========
%% 2. CHEBYSHEV TYPE-I FORMULA
% ==========
Figures
Figure 1
Figure 2
Figure 3
Comparison of IIR Filter Orders
Filter Order
Butterworth
Chebyshev-I
Elliptic
Filter Type
Command Window
New to MATLAB? See resources for Getting Started
Passband Edge = 100.00 Hz
Stopband Edge = 150.00 Hz
Transition Band = 50.00 Hz
>>
Hot days ahead
30°C
Search
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08:39
13-10-2026


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Title
Subtitle
X-Label
Y-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
Text Arrow
FILE
LABELS AND ANNOTATIONS
MATLAB Drive
Files
Name
DSPexp1.m
DSPexp2.m
DSPexp3.m
DSPexp4.m
untitled3.m
untitled4.m
untitled5.m
untitled6.m
untitled7.m
untitled8.m
Workspace
Name
Value
Size
Class
Editor
Figures
Figure 1
Figure 2
Figure 3
Comparison of IIR Filter Phase Responses
Butterworth
Chebyshev Type-I
Elliptic
Phase (degrees)
Frequency (Hz)
Command Window
New to MATLAB? See resources for Getting Started
Stopband Edge = 150.00 Hz
Transition Band = 50.00 Hz
>>
>>
30°C
Sunny
Search
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APPS
FIGURE
FORMAT
TOOLS
Search (Ctrl+Shift+Space)
Priyanka
Save As
Show Code
FILE
Title
Subtitle
X-Label
Y-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
Text Arrow
LABELS AND ANNOTATIONS
/ › MATLAB Drive
Files
Name
DSPexp1.m
DSPexp2.m
DSPexp3.m
DSPexp4.m
untitled3.m
untitled4.m
untitled5.m
untitled6.m
untitled7.m
untitled8.m
Workspace
Name
Value
Size
Class
aB 1×13 double 1×13 double
aC [1,-4.8694,1... 1×7 double
aE [1,-3.0330,3.... 1×5 double
bB 1×13 double 1×13 double
bC [4.6372e-05,... 1×7 double
bE [0.0197,-0.0... 1×5 double
Dev_B_St... 0.0494 1×1 double
Dev_C_Ri... 7.9637e-07 1×1 double
Dev_C_St... 3.3413 1×1 double
Dev_E_St... 2.1975e-08 1×1 double
Editor
Figures
Figure 1
Figure 2
Figure 3
Comparison of IIR Filter Magnitude Responses
Butterworth
Chebyshev Type-I
Elliptic
data1
data2
data3
data4
Fp = 150 Hz
Fst = 150 Hz
Rs = -40 dB
Magnitude (dB)
Frequency (Hz)
Command Window
New to MATLAB? See resources for Getting Started.
Stopband Edge = 150.00 Hz
Transition Band = 50.00 Hz
>>
>>
Search
ENG
IN
09:49
13-10-2026